

Parametric Formula for dy/dx

If all three derivatives exist and $dx/dt \neq 0$,

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}. \quad (1)$$

EXAMPLE 1 Find the tangent to the curve

$$x = \sec t, \quad y = \tan t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2},$$

at the point $(\sqrt{2}, 1)$, where $t = \pi/4$ (Figure 11.12).

Solution The slope of the curve at t is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\sec^2 t}{\sec t \tan t} = \frac{\sec t}{\tan t} \quad \text{Eq. (1)}$$

Setting t equal to $\pi/4$ gives

$$\begin{aligned} \left. \frac{dy}{dx} \right|_{t=\pi/4} &= \frac{\sec(\pi/4)}{\tan(\pi/4)} \\ &= \frac{\sqrt{2}}{1} = \sqrt{2}. \end{aligned}$$

The tangent line is

$$\begin{aligned} y - 1 &= \sqrt{2}(x - \sqrt{2}) \\ y &= \sqrt{2}x - 2 + 1 \\ y &= \sqrt{2}x - 1. \end{aligned}$$

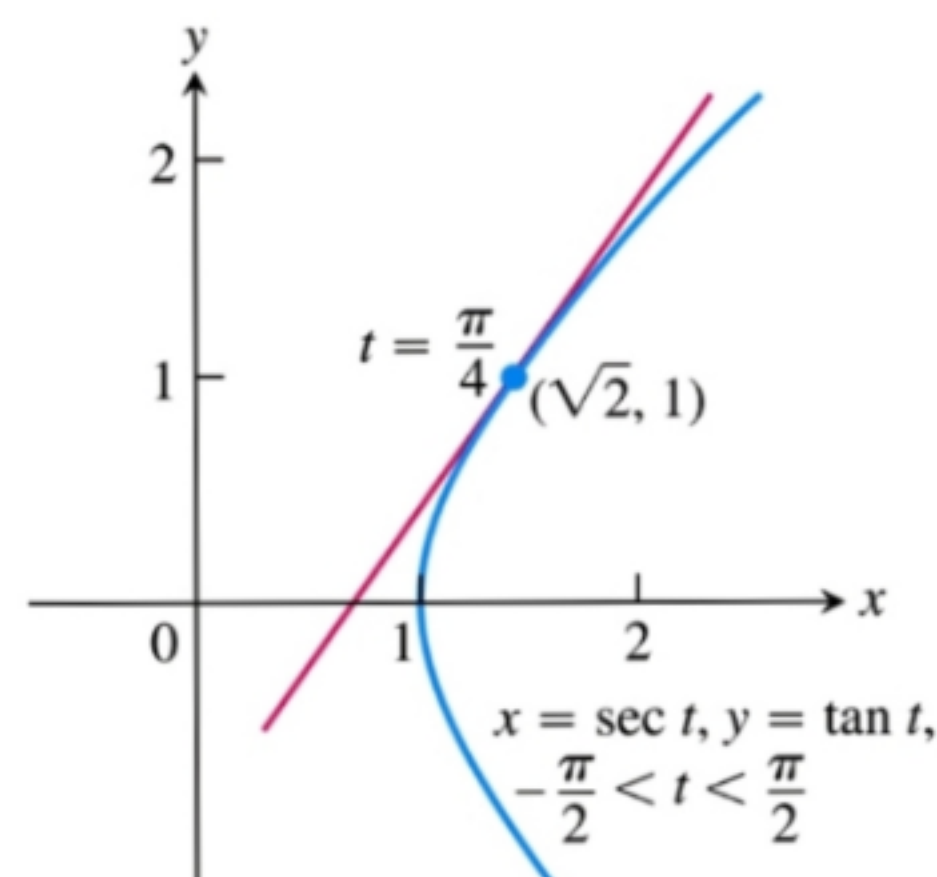


FIGURE 11.12 The curve in Example 1 is the right-hand branch of the hyperbola $x^2 - y^2 = 1$.

Parametric Formula for d^2y/dx^2

If the equations $x = f(t)$, $y = g(t)$ define y as a twice-differentiable function of x , then at any point where $dx/dt \neq 0$ and $y' = dy/dx$,

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt}. \quad (2)$$

EXAMPLE 2 Find d^2y/dx^2 as a function of t if $x = t - t^2$, $y = t - t^3$.

Solution

1. Express $y' = dy/dx$ in terms of t .

$$y' = \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{1 - 3t^2}{1 - 2t}$$

2. Differentiate y' with respect to t .

$$\frac{dy'}{dt} = \frac{d}{dt} \left(\frac{1 - 3t^2}{1 - 2t} \right) = \frac{2 - 6t + 6t^2}{(1 - 2t)^2} \quad \text{Derivative Quotient Rule}$$

3. Divide dy'/dt by dx/dt .

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{(2 - 6t + 6t^2)/(1 - 2t)^2}{1 - 2t} = \frac{2 - 6t + 6t^2}{(1 - 2t)^3} \quad \text{Eq. (2)}$$

DEFINITION If a curve C is defined parametrically by $x = f(t)$ and $y = g(t)$, $a \leq t \leq b$, where f' and g' are continuous and not simultaneously zero on $[a, b]$, and C is traversed exactly once as t increases from $t = a$ to $t = b$, then **the length of C** is the definite integral

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt.$$

EXAMPLE 4 Using the definition, find the length of the circle of radius r defined parametrically by

$$x = r \cos t \quad \text{and} \quad y = r \sin t, \quad 0 \leq t \leq 2\pi.$$

Solution As t varies from 0 to 2π , the circle is traversed exactly once, so the circumference is

$$L = \int_0^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

We find

$$\frac{dx}{dt} = -r \sin t, \quad \frac{dy}{dt} = r \cos t$$

and

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = r^2(\sin^2 t + \cos^2 t) = r^2.$$

So

$$L = \int_0^{2\pi} \sqrt{r^2} dt = r [t]_0^{2\pi} = 2\pi r. \quad \blacksquare$$

Area of Surface of Revolution for Parametrized Curves

If a smooth curve $x = f(t)$, $y = g(t)$, $a \leq t \leq b$, is traversed exactly once as t increases from a to b , then the areas of the surfaces generated by revolving the curve about the coordinate axes are as follows.

1. Revolution about the x -axis ($y \geq 0$):

$$S = \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \quad (5)$$

2. Revolution about the y -axis ($x \geq 0$):

$$S = \int_a^b 2\pi x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \quad (6)$$

EXAMPLE 7 The standard parametrization of the circle of radius 1 centered at the point $(0, 1)$ in the xy -plane is

$$x = \cos t, \quad y = 1 + \sin t, \quad 0 \leq t \leq 2\pi.$$

Use this parametrization to find the area of the surface swept out by revolving the circle about the x -axis (Figure 11.17).

Solution We evaluate the formula

$$\begin{aligned} S &= \int_a^b 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \\ &= \int_0^{2\pi} 2\pi(1 + \sin t) \sqrt{\underbrace{(-\sin t)^2 + (\cos t)^2}_1} dt \\ &= 2\pi \int_0^{2\pi} (1 + \sin t) dt \\ &= 2\pi [t - \cos t]_0^{2\pi} = 4\pi^2. \end{aligned}$$

Eq. (5) for revolution
about the x -axis;
 $y = 1 + \sin t \geq 0$



Tangents to Parametrized Curves

In Exercises 1–14, find an equation for the line tangent to the curve at the point defined by the given value of t . Also, find the value of d^2y/dx^2 at this point.

1. $x = 2 \cos t$, $y = 2 \sin t$, $t = \pi/4$
2. $x = \sin 2\pi t$, $y = \cos 2\pi t$, $t = -1/6$
3. $x = 4 \sin t$, $y = 2 \cos t$, $t = \pi/4$
4. $x = \cos t$, $y = \sqrt{3} \cos t$, $t = 2\pi/3$
5. $x = t$, $y = \sqrt{t}$, $t = 1/4$
6. $x = \sec^2 t - 1$, $y = \tan t$, $t = -\pi/4$
7. $x = \sec t$, $y = \tan t$, $t = \pi/6$
8. $x = -\sqrt{t+1}$, $y = \sqrt{3}t$, $t = 3$
9. $x = 2t^2 + 3$, $y = t^4$, $t = -1$
10. $x = 1/t$, $y = -2 + \ln t$, $t = 1$
11. $x = t - \sin t$, $y = 1 - \cos t$, $t = \pi/3$
12. $x = \cos t$, $y = 1 + \sin t$, $t = \pi/2$
13. $x = \frac{1}{t+1}$, $y = \frac{t}{t-1}$, $t = 2$

Implicitly Defined Parametrizations

Assuming that the equations in Exercises 15–20 define x and y implicitly as differentiable functions $x = f(t)$, $y = g(t)$, find the slope of the curve $x = f(t)$, $y = g(t)$ at the given value of t .

15. $x^3 + 2t^2 = 9$, $2y^3 - 3t^2 = 4$, $t = 2$
16. $x = \sqrt{5 - \sqrt{t}}$, $y(t-1) = \sqrt{t}$, $t = 4$
17. $x + 2x^{3/2} = t^2 + t$, $y\sqrt{t+1} + 2t\sqrt{y} = 4$, $t = 0$

Lengths of Curves

Find the lengths of the curves in Exercises 25–30.

25. $x = \cos t$, $y = t + \sin t$, $0 \leq t \leq \pi$
26. $x = t^3$, $y = 3t^2/2$, $0 \leq t \leq \sqrt{3}$

Surface Area

Find the areas of the surfaces generated by revolving the curves in Exercises 31–34 about the indicated axes.

31. $x = \cos t$, $y = 2 + \sin t$, $0 \leq t \leq 2\pi$; x -axis
32. $x = (2/3)t^{3/2}$, $y = 2\sqrt{t}$, $0 \leq t \leq \sqrt{3}$; y -axis
33. $x = t + \sqrt{2}$, $y = (t^2/2) + \sqrt{2}t$, $-\sqrt{2} \leq t \leq \sqrt{2}$; y -axis
34. $x = \ln(\sec t + \tan t) - \sin t$, $y = \cos t$, $0 \leq t \leq \pi/3$; x -axis